

Measuring the External Field: A Cortical-Proxy Content Observable and the Mean-Field Bound on Media-Driven Opinion Change

Brian Mwai, 1050555, B.Sc. Physics, Department of Natural Sciences, The Catholic University of Eastern Africa. Supervisor: Dr. Songa Mutambi.

Abstract

Physicists model opinion formation with the Ising model, the same mathematics that describes how magnetic spins in a metal align. In its mean-field version, each person holds one of two opinions, is pulled toward the majority by a coupling J , and media acts as an external field h pushing everyone the same way. Studies choose the value of h by hand; it is never measured. This project asks whether it can be measured from the content itself.

Each news article was converted to speech and passed through TRIBE v2, a published AI model from Meta that predicts the brain's response. The project compares predicted activity in regions linked to emotion with regions linked to careful reasoning, giving one number X per article. No person was scanned; every value is a model prediction.

Four hundred articles in four groups (fear-driven, outrage, clickbait and neutral) were measured twice. The groups differ clearly ($\eta^2 = 0.107$, $p < 10^{-8}$), led by fear-driven articles, and the result repeated on the second run. Each group came from a different source, however, so the difference cannot be credited to writing style alone, and a simple word-counting method sorted manipulative from neutral articles better than this measure. The measure is therefore offered as a physical quantity, not a manipulation detector.

Writing the field as $h = \alpha X$, the strength α with which content pushes opinion could not be pinned down from the data, so no value is claimed. Instead, the thesis derives a condition that needs no fitted value: in this model, media can flip a population's majority only if $\alpha \geq h_c(\beta J) / \Delta X$, where h_c is the critical field and ΔX the spread of the measure. With the measured $\Delta X = 0.124$, this requires $\alpha \geq 4.29$ at $\beta J = 2$, a minimum any future media-influence model of this kind must meet.

Keywords: opinion dynamics, mean-field Ising model, external field, media influence, brain-response prediction.